INDUCTION PRINCIPLE FOR QUASI-COMPACT OPENS, SEPARATED VARIANT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the counted statement is the author lemma environment labelled lemma-induction-principle-separated; the proof environment immediately following it is its own proof; the template argument cited by the proof is delivered as its own card (lemma-induction-principle)

Editorial note. The label lemma-induction-principle-separated names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the induction principle template (lemma-induction-principle of the same chapter, delivered as its own corpus card): a property P of quasi-compact quasi-separated etale opens holds on all of B if it holds on affines, passes to quasi-compact opens, and is inherited along elementary distinguished squares
- the observation that in hypothesis (4)(d) the fiber product U x_W V is a quasi-compact open of the affine V, hence lies in B by conditions (2) and (3)

Source reference. Original reference 08DR: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1563–1563. Prerequisite pointer invoked at 1670; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let X be a quasi-compact and quasi-separated algebraic space over S. Let $\mathcal{B} \subset \mathrm{Ob}(X_{spaces,étale})$. Let P be a property of the elements of $\mathcal{B}$. Assume that*

(1) *every $W \in \mathcal{B}$ is quasi-compact and quasi-separated,*
(2) *if $W \in \mathcal{B}$ and $U \subset W$ is quasi-compact open, then $U \in \mathcal{B}$,*
(3) *if $V \in \mathrm{Ob}(X_{spaces,étale})$ is affine, then (a) $V \in \mathcal{B}$ and (b) P holds for V,*
(4) *for every elementary distinguished square $(U \subset W, f : V \to W)$ such that*
   (a) *$W \in \mathcal{B}$,*
   (b) *U is quasi-compact,*
   (c) *V is affine, and*
   (d) *P holds for U, V, and $U \times_W V$,*

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then P holds for W.

Then P holds for every $W \in \mathcal{B}$.

Proof. This is proved in exactly the same manner as the proof of Lemma 08DR. (We remark that (4)(d) makes sense as $U \times_W V$ is a quasi-compact open of V hence an element of $\mathcal{B}$ by conditions (2) and (3).) □